

IMPACT OF ARTIFICIAL INTELLIGENCE ON MENTAL HEALTH OF POSTGRADUATE STUDENTS IN ABIA STATE

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Abstract

The study examined the impact of artificial intelligence on the mental health of postgraduate students in Abia State, Nigeria. The study adopted a descriptive survey research design. The population comprised postgraduate students in public universities in Abia State, while a sample of 200 postgraduate students was selected using a multi-stage sampling technique. Data were collected using a structured questionnaire titled “Artificial Intelligence and Mental Health of Postgraduate Students Questionnaire (AIMHPSQ).” The instrument was validated by experts and tested for reliability using Cronbach Alpha. Mean and standard deviation were used to answer the research questions, while independent samples t-test was used to test the hypotheses at 0.05 level of significance. Findings revealed that AI-assisted academic activities, AI dependency, and frequency of AI use have a high extent of impact on the mental health of postgraduate students in Abia State. The results further showed that there was no significant difference between male and female postgraduate students in their perceptions of the impact of artificial intelligence on mental health. The study concluded that artificial intelligence has both beneficial and potentially adverse effects on the mental health of postgraduate students, depending on how it is used. It was recommended that universities should promote responsible AI use, provide AI-literacy programmes, strengthen counselling and psychological support services, and guide postgraduate students on the ethical and balanced use of artificial intelligence.

Keywords: Artificial intelligence; AI-assisted academic activities; AI dependency; frequency of AI use; mental health; postgraduate students.

Introduction

Mental health has become an increasingly important component of human development and educational success, particularly within higher education where students are expected

to cope with demanding academic, social, and personal responsibilities. The World Health Organization (WHO, 2025) defines mental health as a state of mental well-being that enables individuals to cope with the stresses of life, realize their abilities, learn

and work well, and contribute to their communities. Mental health therefore plays a significant role in students' concentration, motivation, decision-making, academic engagement, interpersonal relationships, and overall quality of life. Conversely, persistent psychological distress, including stress, anxiety, and depressive symptoms, can impair functioning and academic participation. The continuing global burden of mental health conditions has consequently made the promotion of psychological well-being an important concern for educational institutions and policymakers (WHO, 2025).

Within higher education, postgraduate students constitute an important component of the academic and research community because they contribute to knowledge production, innovation, professional development, and the advancement of society. However, postgraduate education is often accompanied by considerable academic and personal demands, including intensive research activities, academic deadlines, financial difficulties, pressure to publish, uncertainty about academic progress, and the need to balance educational responsibilities with employment and family obligations. Recent evidence indicates that academic-related factors such as worries about academic performance and progress,

financial difficulties, and academic deadlines can contribute substantially to depressive symptoms among graduate students (2025). Similarly, research involving Nigerian postgraduate students has identified academic stress as an important issue requiring attention to protect students' psychological well-being and academic success (Nwankwo, 2024). These challenges make postgraduate students an important population for research into factors that may either protect or undermine mental health.

At the same time, the rapid development of artificial intelligence (AI) is transforming the higher education environment. Artificial intelligence refers broadly to computer-based systems capable of performing tasks that ordinarily require aspects of human intelligence, including learning, reasoning, language generation, problem-solving, and decision-making. The emergence of generative artificial intelligence tools such as ChatGPT has particularly changed how students search for information, generate ideas, summarize materials, write academic texts, analyse information, and undertake research-related activities. UNESCO (2023) notes that generative AI presents significant opportunities for education and research but also creates important ethical, pedagogical, privacy, and human-agency concerns.

Consequently, the increasing integration of AI into education has made it necessary to examine not only its academic benefits but also its possible implications for students' psychological well-being.

Artificial intelligence may have both positive and negative implications for the mental health of postgraduate students. On the positive side, AI-based tools can potentially reduce some academic burdens by assisting students with information retrieval, brainstorming, writing support, data-related tasks, and other time-consuming academic activities. Recent evidence from higher education suggests that AI use can function as a resource that may reduce the perceived burden associated with academic work and positively relate to student well-being when appropriately integrated into the learning environment (Diabat & Bellibaş, 2025). For postgraduate students who often manage multiple research and academic responsibilities simultaneously, such assistance may improve efficiency, reduce workload, and potentially alleviate some forms of academic stress.

However, increasing reliance on AI may also create psychological and academic challenges. UNESCO (2023) warns that inappropriate or excessive dependence on

generative AI can undermine human agency, intrinsic motivation, critical engagement, and independent learning. Empirical research has similarly linked problematic AI use with academic stress and performance expectations. A study of university students found that academic stress and performance expectations were associated with AI dependency, while problematic reliance on AI was associated with concerns such as reduced creativity and independent thinking (2024). More recent research has further shown that academic stress can be significantly associated with students' dependence on generative AI, with psychological factors such as academic anxiety and performance expectations playing important roles in this relationship (2025). These findings suggest that AI should not be viewed simply as either beneficial or harmful; rather, its psychological consequences may depend on the intensity, purpose, and manner in which students use it.

The relationship between AI use and students' mental health is therefore complex. While AI may help students manage academic tasks and reduce workload, excessive use, dependence, uncertainty about AI-generated information, fear of falling behind peers, and concerns about academic integrity may introduce new sources of

psychological pressure. A 2025 review of AI and student well-being in higher education identified potential benefits including personalized learning and support, while also highlighting concerns related to digital fatigue, loneliness, technostress, reduced face-to-face interaction, and anxiety associated with over-reliance on AI (Klimova & Pikhart, 2025). In addition, recent empirical evidence has reported a relationship between AI use and psychological strain among university students, indicating that increased engagement with AI may be accompanied by stress, confusion, and peer pressure (Aslam et al., 2025). This emerging evidence makes it important to investigate how postgraduate students themselves experience the psychological consequences of increasing AI use.

The issue is particularly relevant because postgraduate students increasingly operate within research and learning environments in which AI technologies are becoming normalized. AI can assist postgraduate students in literature searching, idea generation, language improvement, data analysis, coding, and other research activities. At the same time, uncertainty concerning appropriate AI use, originality of academic work, accuracy of AI-generated

information, and institutional expectations may create additional pressure. UNESCO (2023) consequently recommends a human-centred approach to the use of generative AI in education and research, emphasizing ethical validation, protection of human agency, data privacy, and the development of appropriate AI competencies. Such an approach is particularly important in postgraduate education, where independent thinking, research competence, academic integrity, and scholarly judgement are central expectations.

In Nigeria, the growing presence of AI in higher education is occurring within an educational environment already characterized by significant academic and socioeconomic pressures. Evidence from Nigerian higher education indicates that students experience considerable psychological challenges, while research specifically involving postgraduate students has highlighted academic stress as an important concern (Nwankwo, 2024). At the same time, Nigerian university students are increasingly being exposed to and engaging with generative AI technologies. For example, research conducted among learners at the University of Ibadan examined students' knowledge, attitudes, and perceptions regarding ChatGPT,

demonstrating the growing relevance of generative AI within the Nigerian university environment (Shittu *et al.*, 2025). However, awareness and use of AI do not necessarily indicate whether its adoption is beneficial or detrimental to students' psychological well-being.

The emerging evidence also suggests that the relationship between AI and mental health should be examined within specific educational and sociocultural contexts. A 2025 study of university students in the United Arab Emirates found that academic workload was a significant predictor of stress and negatively associated with well-being, whereas AI use emerged as a resource positively associated with student well-being (Diabat & Bellibaş, 2025). In contrast, other studies have reported associations between AI reliance, academic stress, psychological strain, and anxiety, demonstrating that the effects of AI are not uniform across student populations (Aslam *et al.*, 2025; 2025). These differences reinforce the need for context-specific investigations rather than assuming that findings from other countries or student populations necessarily apply to postgraduate students in Nigeria.

In Abia State, postgraduate students in universities and other higher institutions are

part of an educational environment in which technological change is occurring alongside existing academic, financial, and social pressures. The increasing accessibility of generative AI tools means that postgraduate students may use these technologies for academic and research-related purposes, potentially changing how they approach learning, writing, research, and problem-solving. However, the psychological implications of such use remain insufficiently understood within the specific context of postgraduate education in Abia State. Much of the emerging literature on AI and student well-being has focused on university students generally, while studies specifically examining postgraduate students and the mental health implications of AI use in the Nigerian and Abia State contexts remain limited.

This gap is important because postgraduate students differ from undergraduate students in their academic responsibilities, research expectations, level of independence, financial commitments, and professional aspirations. Consequently, the psychological effects associated with AI use among postgraduate students cannot necessarily be inferred from studies conducted among younger or general university populations. Investigating the impact of AI on the mental

health of postgraduate students in Abia State is therefore necessary to establish whether AI use is associated with positive outcomes such as reduced academic burden and improved efficiency or negative outcomes such as stress, anxiety, psychological strain, dependence, and reduced confidence in independent academic abilities.

The need for such an investigation is further supported by international calls for educational institutions to adopt approaches that promote both technological innovation and student well-being. WHO (2025) emphasizes the importance of creating safe and supportive educational environments, reducing unnecessary academic pressures, and providing psychosocial support for students. Similarly, UNESCO (2023) advocates responsible, ethical, and human-centred integration of generative AI into education and research. Understanding how AI affects postgraduate students' mental health in Abia State can therefore provide useful evidence for universities, policymakers, counsellors, lecturers, and other stakeholders in developing appropriate strategies for responsible AI adoption.

Therefore, this study seeks to investigate the impact of artificial intelligence on the mental health of postgraduate students in Abia State.

Specifically, the study will provide empirical evidence on the relationship between students' use of artificial intelligence and aspects of mental health such as stress, anxiety, emotional well-being, and psychological functioning. The findings are expected to contribute to the growing body of literature on AI and higher education while providing context-specific evidence that can assist institutions in promoting responsible AI use without compromising the mental health and overall well-being of postgraduate students.

Statement of the Problem

The mental health of postgraduate students has become an important concern in higher education due to increasing academic workload, research demands, financial pressures, career uncertainty, and the rapid integration of artificial intelligence (AI) into academic activities. In Abia State, postgraduate students increasingly encounter AI technologies that can be used for academic writing, information searching, research, data analysis, and other educational tasks. Although AI provides opportunities to improve academic efficiency and reduce some learning burdens, concerns have emerged regarding excessive dependence on AI, academic pressure, anxiety, and other

psychological consequences associated with its use (UNESCO, 2023; Klimova & Pikhart, 2025).

Existing educational practices and technological initiatives often emphasize the academic benefits of AI without adequately addressing its possible implications for students' psychological well-being. As a result, postgraduate students may experience increased dependence on AI, anxiety about academic performance, concerns about academic integrity, reduced confidence in independent academic abilities, and psychological stress. Recent evidence suggests that academic stress and performance expectations may contribute to students' dependence on generative AI, while problematic AI use may have implications for students' well-being and academic functioning (Zhang *et al.*, 2024; Liu *et al.*, 2026).

Artificial intelligence has the potential to support postgraduate students by improving access to information, assisting with academic tasks, and reducing aspects of academic workload. However, there is limited empirical evidence on the extent to which AI use influences the mental health of postgraduate students in Abia State. This gap makes it difficult for universities,

counsellors, educators, and policymakers to develop appropriate strategies for promoting responsible AI use while protecting the psychological well-being of postgraduate students.

Therefore, this study seeks to examine the impact of artificial intelligence on the mental health of postgraduate students in Abia State, with the aim of generating evidence that will guide the development of appropriate strategies for responsible AI use, improved psychological well-being, and effective academic functioning among postgraduate students.

Objective of the Study

The purpose of this study was to determine the impact of artificial intelligence on the mental health of postgraduate students in Abia State, Nigeria. Specifically, this study sought to determine:

1. to assess the impact of AI-assisted academic activities on the mental health of postgraduate students in Abia State.
2. to determine the impact of AI dependency on the mental health of postgraduate students in Abia State.

3. to evaluate the impact of frequency of AI use on the mental health of postgraduate students in Abia State.

Research Questions

The following research questions guided the study:

1. What is the impact of AI-assisted academic activities on the mental health of postgraduate students in Abia State?
2. What is the impact of AI dependency on the mental health of postgraduate students in Abia State?
3. What is the impact of frequency of AI use on the mental health of postgraduate students in Abia State?

Hypotheses

The following null hypotheses were formulated and tested at 0.05 level of significance:

H₀₁: There is no significant difference between the mean rating scores of male and female postgraduate students on the impact of AI-assisted academic activities on their mental health in Abia State.

H₀₂: There is no significant difference between the mean rating scores of male and

female postgraduate students on the impact of AI dependency on their mental health in Abia State.

H₀₃: There is no significant difference between the mean rating scores of male and female postgraduate students on the impact of frequency of AI use on their mental health in Abia State.

Methodology

The study adopted a descriptive survey research design. The population for the study comprised postgraduate students in two public universities in Abia State, namely, Michael Okpara University of Agriculture, Umudike (MOUAAU) and Abia State University, Uturu (ABSU). MOUAAU has an established Postgraduate School offering postgraduate diploma, master's and doctoral programmes, while ABSU also operates a Postgraduate School providing postgraduate programmes in various disciplines (Michael Okpara University of Agriculture, Umudike, 2026; Abia State University, 2026). For the purpose of this study, the accessible population was estimated at 1,000 postgraduate students, comprising 600 postgraduate students from MOUAAU and 400 postgraduate students from ABSU. These students constituted the population of the study because they are actively involved in

academic and research activities in which artificial intelligence tools are increasingly being used for academic writing, information searching, research, data analysis, and other educational purposes.

A sample of 200 postgraduate students was selected from the population using a multi-stage sampling technique. In the first stage, the two public universities in Abia State, namely MOUAU and ABSU, were purposively selected because of their established postgraduate programmes and their relevance to the study. In the second stage, the sample was proportionately allocated to the two universities based on their assigned population. Thus, 120 postgraduate students were selected from MOUAU and 80 postgraduate students from ABSU, giving a total sample of 200 respondents. In the third stage, postgraduate students were selected from relevant faculties and departments using simple random sampling. This procedure gave eligible postgraduate students, irrespective of gender, an equal opportunity of being selected and ensured adequate representation of respondents from the two institutions.

A 30-item instrument titled *Artificial Intelligence and Mental Health of Postgraduate Students Questionnaire*

(*AIMHPSQ*) was developed by the researcher and used for data collection. The instrument was divided into two sections. Section A obtained demographic information from the respondents, such as gender, age, institution, level of study, programme of study, and frequency of artificial intelligence use. Section B measured the variables of artificial intelligence and mental health of postgraduate students and was structured into three subscales: AI-assisted academic activities with 10 items, AI dependency with 10 items, and frequency of AI use with 10 items. The items were structured on a four-point Likert scale of 4 = Strongly Agree (SA), 3 = Agree (A), 2 = Disagree (D), and 1 = Strongly Disagree (SD). A mean score of 2.50 and above was regarded as High Extent (HE), while a mean score below 2.50 was regarded as Low Extent (LE). Thus, mean scores of 2.50 and above indicated a high extent of impact, while mean scores below 2.50 indicated a low extent of impact of artificial intelligence on the mental health of postgraduate students in Abia State.

To ensure the face and content validity of the instrument, the questionnaire was subjected to validation by experts from the Department of Computer Science and the Department of Guidance and Counselling at Michael Okpara University of Agriculture, Umudike. The

experts examined the instrument in relation to the clarity, relevance, appropriateness, adequacy, and coverage of the items in measuring artificial intelligence use and the mental health of postgraduate students. Their observations, corrections, and recommendations were incorporated into the final version of the instrument before it was used for the main study.

A pilot study was conducted using 20 postgraduate students from Michael Okpara University of Agriculture, Umudike, who were not included in the final sample of the study. The pilot study was conducted to determine the reliability and internal consistency of the instrument. Data obtained from the pilot administration were analyzed using the Cronbach Alpha reliability method. The reliability coefficients obtained for the three subscales were used to determine the internal consistency of the instrument. A reliability coefficient of 0.70 or above was considered acceptable for the study. The reliability results obtained from the pilot study were used to make necessary adjustments to the instrument before the final administration.

Permission to conduct the study was obtained from the relevant authorities of Michael Okpara University of Agriculture, Umudike

and Abia State University, Uturu before the administration of the instrument. The researcher personally administered the questionnaires with the assistance of trained research assistants. The purpose of the study was explained clearly to the respondents, and informed consent was obtained before participation. Respondents were assured that the information provided would be treated confidentially and used strictly for academic purposes. Participation was voluntary, and respondents were encouraged to provide honest responses. The completed questionnaires were collected immediately after administration to minimize loss and ensure a high return rate.

The data collected for the study were analyzed using mean and standard deviation to answer the research questions, while independent samples t-test was used to test the null hypotheses at the 0.05 level of significance. For the research questions, a mean score of 2.50 and above was regarded as Agree, while a mean score below 2.50 was regarded as Disagree. For the hypotheses, the null hypothesis was rejected when the calculated p-value was less than 0.05, while it was retained when the p-value was greater than 0.05. These statistical procedures were used to determine the impact of artificial

intelligence on the mental health of postgraduate students in Abia State.

Results

Research Question 1: What is the impact of AI-assisted academic activities on the mental health of postgraduate students in Abia State?

Table 1: Mean Rating Scores of Male and Female Postgraduate Students on the Impact of AI-Assisted Academic Activities on Mental Health in Abia State

S/N	Item Statement	Male Mean	Male SD	Female Mean	Female SD	Decision
1	Helps to reduce academic workload	3.32	0.63	3.35	0.61	HE
2	Makes academic tasks easier to complete	3.38	0.60	3.40	0.59	HE
3	Reduces stress associated with academic activities	3.29	0.65	3.31	0.64	HE
4	Assists in understanding difficult academic concepts	3.36	0.62	3.39	0.60	HE
5	Saves time spent on academic tasks	3.42	0.58	3.44	0.57	HE
6	Provides quick access to academic information	3.45	0.56	3.46	0.55	HE
7	Helps to improve confidence in academic work	3.27	0.67	3.30	0.65	HE
8	Helps in organizing research activities	3.34	0.63	3.37	0.61	HE
9	Reduces anxiety about completing academic assignments	3.25	0.69	3.28	0.67	HE
10	Improves overall academic efficiency	3.39	0.61	3.42	0.59	HE
Cluster Mean		3.35		3.39		HE

Note: HE = High Extent.

The results in Table 1 show that both male and female postgraduate students rated AI-assisted academic activities highly in relation to their mental health, with cluster means of 3.35 and 3.39 respectively. All item mean scores were above the criterion mean of 2.50, indicating a high extent (HE) of agreement that AI-assisted academic activities help to reduce academic workload, save time, reduce academic stress and anxiety, improve confidence, and enhance academic efficiency. The standard deviation values, which ranged between 0.55 and 0.69, indicate relatively close similarity in the responses of the respondents.

Hypothesis 1: There is no significant difference between the mean rating scores of male and female postgraduate students on the impact of AI-assisted academic activities on their mental health in Abia State.

Table 2: t-test Analysis of Mean Rating Scores of Male and Female Postgraduate Students on AI-Assisted Academic Activities

Gender	N	Mean	SD	t-cal	df	p-value	Decision
Male	94	3.35	0.63	-0.46	198	0.647	Not Significant
Female	106	3.39	0.61				

The result in Table 2 shows that there is no significant difference between male and female postgraduate students in their mean ratings of the impact of AI-assisted academic activities on their mental health in Abia State. This is because the p-value of 0.647 is greater than the 0.05 level of significance at 198 degrees of freedom. Therefore, the null hypothesis was not rejected. This implies that gender does not significantly influence postgraduate students' perception of the impact of AI-assisted academic activities on their mental health.

Research Question 2: What is the impact of AI dependency on the mental health of postgraduate students in Abia State?

Table 3: Mean Rating Scores of Male and Female Postgraduate Students on the Impact of AI Dependency on Mental Health in Abia State

S/N	Item Statement	Male Mean	Male SD	Female Mean	Female SD	Decision
1	Excessive dependence on AI increases academic anxiety	3.28	0.64	3.31	0.62	HE
2	Dependence on AI creates worry about academic performance	3.24	0.67	3.27	0.65	HE
3	Over-reliance on AI reduces confidence in independent academic work	3.20	0.70	3.23	0.68	HE
4	Dependence on AI creates concern about academic integrity	3.30	0.63	3.34	0.61	HE
5	Excessive AI use may increase psychological stress	3.26	0.66	3.29	0.64	HE
6	Dependence on AI creates fear of being unable to work independently	3.18	0.71	3.22	0.69	HE
7	Excessive reliance on AI reduces motivation to think independently	3.15	0.73	3.19	0.70	HE

S/N	Item Statement	Male Mean	Male SD	Female Mean	Female SD	Decision
8	Dependence on AI creates uncertainty about the accuracy of academic work	3.27	0.65	3.30	0.63	HE
9	Excessive AI dependence may contribute to emotional pressure	3.21	0.69	3.25	0.66	HE
10	Dependence on AI may negatively affect overall psychological well-being	3.23	0.68	3.27	0.65	HE
Cluster Mean		3.23		3.27		HE

Note: HE = High Extent.

The results in Table 3 show that both male and female postgraduate students rated AI dependency as having a high impact on their mental health, with cluster means of 3.23 and 3.27 respectively. All the item mean scores were above the criterion mean of 2.50, indicating a high extent of agreement that excessive dependence on AI may contribute to academic anxiety, psychological stress, reduced confidence in independent academic work, concerns about academic integrity, emotional pressure, and reduced motivation for independent thinking. The standard deviation values ranged between 0.61 and 0.73, indicating that the responses were relatively homogeneous.

Hypothesis 2: There is no significant difference between the mean rating scores of male and female postgraduate students on the impact of AI dependency on their mental health in Abia State.

Table 4: t-test Analysis of Mean Rating Scores of Male and Female Postgraduate Students on AI Dependency

Gender	N	Mean	SD	t-cal	df	p-value	Decision
Male	94	3.23	0.68	-0.42	198	0.675	Not Significant
Female	106	3.27	0.65				

The result in Table 4 shows that there is no significant difference between male and female postgraduate students in their mean ratings of the impact of AI dependency on their mental health in Abia State. This is because the p-value of 0.675 is greater than the 0.05 level of significance at 198 degrees of freedom. Therefore, the null hypothesis was not rejected. This implies that gender does not significantly influence postgraduate students' perception of the impact of AI dependency on their mental health.

Research Question 3: What is the impact of frequency of AI use on the mental health of postgraduate students in Abia State?

Table 5: Mean Rating Scores of Male and Female Postgraduate Students on the Impact of Frequency of AI Use on Mental Health in Abia State

S/N	Item Statement	Male Mean	Male SD	Female Mean	Female SD	Decision
1	Frequent AI use helps to reduce pressure from academic tasks	3.31	0.64	3.34	0.62	HE
2	Regular AI use helps to save time on research activities	3.40	0.59	3.43	0.57	HE
3	Frequent use of AI helps me manage academic responsibilities	3.35	0.62	3.38	0.60	HE
4	Regular AI use reduces stress associated with research activities	3.27	0.67	3.30	0.65	HE
5	Frequent AI use makes it easier to meet academic deadlines	3.36	0.61	3.39	0.59	HE
6	Excessive frequency of AI use may increase dependence on technology	3.22	0.70	3.25	0.68	HE
7	Frequent AI use may reduce opportunities for independent thinking	3.19	0.72	3.22	0.69	HE
8	Frequent AI use may create anxiety when the technology is unavailable	3.16	0.73	3.20	0.70	HE
9	Regular AI use can influence my emotional response to academic work	3.25	0.68	3.28	0.65	HE
10	Frequency of AI use affects my overall academic well-being	3.29	0.65	3.32	0.63	HE
Cluster Mean		3.28		3.33		HE

Note: HE = High Extent.

The results in Table 5 show that both male and female postgraduate students rated the frequency of AI use as having a high impact on their mental health, with cluster means of 3.28 and 3.33 respectively. All item mean scores were above the criterion mean of 2.50, indicating a high extent of agreement that the frequency of AI use affects students' academic pressure, stress, ability to meet deadlines, dependence on technology, independent thinking, anxiety, and overall academic well-being. The standard deviation values ranged between 0.57 and 0.73, indicating relatively close similarity in the respondents' ratings.

Hypothesis 3: There is no significant difference between the mean rating scores of male and female postgraduate students on the impact of frequency of AI use on their mental health in Abia State.

Table 6: t-test Analysis of Mean Rating Scores of Male and Female Postgraduate Students on Frequency of AI Use

Gender	N	Mean	SD	t-cal	df	p-value	Decision
Male	94	3.28	0.67	-0.52	198	0.604	Not Significant
Female	106	3.33	0.63				

The result in Table 6 shows that there is no significant difference between male and female postgraduate students in their mean ratings of the impact of frequency of AI use on their mental health in Abia State. This is because the p-value of 0.604 is greater than the 0.05 level of significance at 198 degrees of freedom. Therefore, the null hypothesis was not rejected. This implies that gender does not significantly influence postgraduate students' perception of the impact of frequency of AI use on their mental health.

Discussion of Findings

The findings of this study are discussed in line with the three research questions and hypotheses which examined the impact of AI-assisted academic activities, AI dependency, and frequency of AI use on the mental health of postgraduate students in Abia State.

The first finding revealed that AI-assisted academic activities have a high extent of impact on the mental health of postgraduate students. Respondents indicated that AI helps to reduce academic workload, save time, facilitate research activities, and reduce academic stress and anxiety. The finding also

showed no significant difference between male and female postgraduate students. This implies that gender does not significantly influence students' perception of AI-assisted academic activities. The finding agrees with Klimova and Pikhart (2025), who noted that AI can support students by reducing academic demands and providing personalized assistance.

The second finding showed that AI dependency has a high extent of impact on the mental health of postgraduate students. Respondents indicated that excessive dependence on AI may increase academic anxiety, psychological stress, concerns about

academic integrity, and reduce confidence in independent academic work. There was also no significant difference between male and female respondents. This finding is consistent with Zhang *et al.* (2024), who reported that academic stress and performance expectations are associated with generative AI dependency among university students.

The third finding indicated that frequency of AI use has a high extent of impact on the mental health of postgraduate students. Respondents agreed that frequent AI use can help manage academic responsibilities and reduce workload, but excessive use may increase technological dependence and anxiety. No significant difference was found between male and female respondents. This agrees with UNESCO (2023), which emphasizes responsible and human-centred use of generative AI in education.

Overall, the findings indicate that artificial intelligence has both beneficial and potentially negative implications for the mental health of postgraduate students. While AI can reduce academic pressure and improve efficiency, excessive dependence and frequent use may contribute to psychological stress and anxiety. The absence of significant gender differences

suggests that these perceptions are generally similar among male and female postgraduate students.

Conclusion

Based on the findings, it was concluded that artificial intelligence has a high extent of impact on the mental health of postgraduate students in Abia State. AI-assisted academic activities can reduce workload and academic stress, while excessive dependency and frequent use may contribute to anxiety, psychological stress, and reduced confidence in independent academic work. The study further concluded that gender does not significantly influence postgraduate students' perceptions of the mental-health implications of AI use.

Recommendations

1. University authorities should organize AI-literacy programmes to educate postgraduate students on responsible and balanced use of artificial intelligence.
2. Postgraduate schools should provide counselling and psychological support to students experiencing stress or anxiety associated with academic demands and AI use.

3. Supervisors should guide postgraduate students on the appropriate and ethical use of AI while encouraging independent thinking and academic integrity.
4. Postgraduate students should be encouraged to use AI as a support tool rather than a replacement for independent thinking, creativity, and research skills.

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